

Initial Project Planning Template

Date	27 June 2026
Team ID	
Project Name	Rising water
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	ML Pipeline Development	USN-1	As a developer, I will collect historical rainfall and meteorological data to form the training dataset.	2	High	Pandiri Gireesh	25 June 2026	25 June 2026
Sprint-1	ML Pipeline Development	USN-2	As a developer, I will handle missing values, cap outliers using IQR, and scale features using StandardScaler.	2	High	Pandiri Gireesh	25 June 2026	25 June 2026
Sprint-1	ML Pipeline Development	USN-3	As a developer, I will train multiple classification models (Logistic Regression, Decision Tree, Random Forest, XGBoost) and evaluate their accuracy.	3	High	Pandiri Gireesh	25 June 2026	25 June 2026
Sprint-1	ML Pipeline Development	USN-4	As a developer, I will serialize the best performing model (XGBoost) and the preprocessor using joblib for backend deployment.	1	Medium	Pandiri Gireesh	25 June 2026	25 June 2026

Sprint-2	Web App (Backend & UI)	USN-5	As a user, I can register and login securely to the application so that my prediction history is saved.	3	High	Pandiri Gireesh	26 June 2026	26 June 2026
Sprint-2	Web App (Backend & UI)	USN-6	As a system, I will connect to a PostgreSQL database using SQLAlchemy to store users and prediction logs.	2	High	Pandiri Gireesh	26 June 2026	26 June 2026
Sprint-2	Web App (Backend & UI)	USN-7	As a user, I can interact with a responsive UI to input weather data and receive instant flood predictions.	3	High	Pandiri Gireesh	26 June 2026	26 June 2026
Sprint-2	Web App (Backend & UI)	USN-8	As a user, I can view detailed reports (ROC curves, confusion matrix) and my past prediction history via a dashboard.	2	Medium	Pandiri Gireesh	26 June 2026	26 June 2026
Sprint-3	Deployment & Testing	USN-9	As a developer, I will deploy the application to Render and configure environment variables securely.	2	High	Pandiri Gireesh	27 June 2026	27 June 2026
Sprint-3	Deployment & Testing	USN-10	As a developer, I will run performance tests using JMeter and Locust to ensure the application handles concurrent user loads.	2	Low	Pandiri Gireesh	27 June 2026	27 June 2026